

5 (a) State Kirchhoff's first law.

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 [1]

(b) Fig. 5.1 shows a circuit containing a thermistor T that has a negative temperature coefficient.

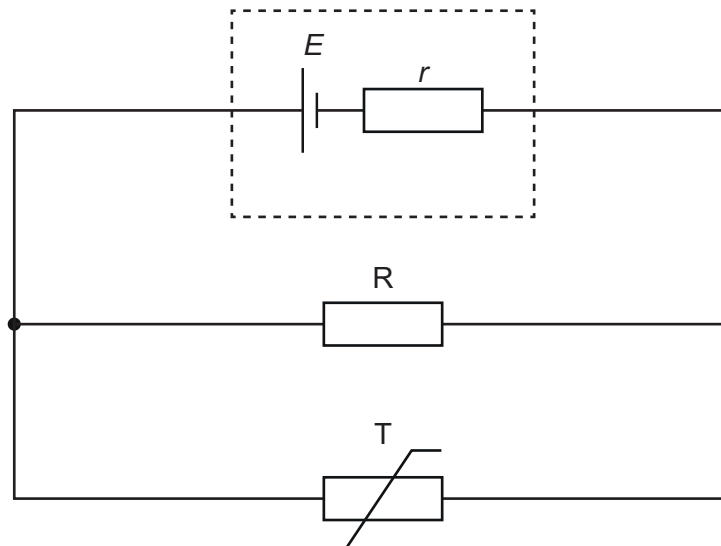


Fig. 5.1

(i) The thermistor has resistance R_0 at a temperature of 0°C .

On Fig. 5.2, sketch a possible variation of the resistance of the thermistor with temperature between 0°C and 100°C .

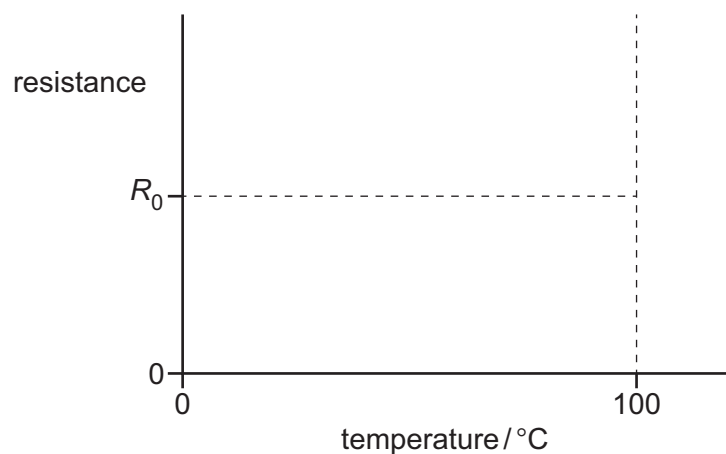


Fig. 5.2

[2]



- (ii) With reference to the current in the cell, explain why the current in **resistor R** decreases with increasing temperature of the thermistor.

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..... [3]

- (c) The electromotive force (e.m.f.) E of the cell in Fig. 5.1 is 1.50 V. The internal resistance r of the cell is $0.12\ \Omega$.

Resistor R has a resistance of $6.00\ \Omega$.

At a particular temperature of the thermistor, the current in R is 0.200 A.

For this temperature of the thermistor, determine:

- (i) the current in the cell

current = A [2]

- (ii) the resistance of the thermistor.

resistance = Ω [2]

[Total: 10]

